

Queries and suspected misprints

PRINCIPLES OF CONTINUUM MECHANICS

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Reader's notes

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The following notes were accumulated during a line-by-line reading of Chapters 1–12. They are grouped by confidence: Part I contains items I believe to be genuine misprints, verified by independent computation; Part II contains statements that appear to be stronger or looser than intended; Part III contains notational and pedagogical points where a sentence of clarification would help the reader; Part IV concerns the figures. Equation numbers and page numbers refer to the printed text.

Part I. Verified misprints

1. Equation (2.32), p. 45. The printed spatial time derivatives are

$$\tilde{v}'_2 = \frac{1 - 2t^2}{1 + t^2} x_2 - \frac{2t}{1 + t^2} x_3, \quad \tilde{v}'_3 = \frac{1 - 2t^2}{1 + t^2} x_3 - \frac{2t}{1 + t^2} x_2.$$

From the inversion (2.31) the spatial velocity components are $\tilde{v}_2 = (1 + t^2)^{-1}(tx_2 + x_3)$ and $\tilde{v}_3 = (1 + t^2)^{-1}(tx_3 - x_2)$, whence, by the quotient rule at fixed $\underline{x}$,

$$\tilde{v}'_2 = \frac{(1 + t^2) - t(2t)}{(1 + t^2)^2} x_2 - \frac{2t}{(1 + t^2)^2} x_3 = \frac{1 - t^2}{(1 + t^2)^2} x_2 - \frac{2t}{(1 + t^2)^2} x_3,$$

and similarly for $\tilde{v}'_3$. Both the numerator ($1 - t^2$, not $1 - 2t^2$) and the exponent of the denominator differ from the printed values.

That the corrected expressions are the right ones is confirmed by the consistency check the text itself proposes. Using (2.29) with the printed matrix (2.33), which is correct,

$$\begin{aligned} \tilde{a}_2 &= \tilde{v}'_2 + \tilde{v}_{2,2}\tilde{v}_2 + \tilde{v}_{2,3}\tilde{v}_3 \\ &= \frac{(1 - t^2)x_2 - 2tx_3}{(1 + t^2)^2} + \frac{t}{1 + t^2} \cdot \frac{tx_2 + x_3}{1 + t^2} + \frac{1}{1 + t^2} \cdot \frac{tx_3 - x_2}{1 + t^2} \\ &= \frac{(1 - t^2 + t^2 - 1)x_2 + (-2t + t + t)x_3}{(1 + t^2)^2} = 0, \end{aligned}$$

in agreement with the stated result $\tilde{a}_2 = 0$. With the printed form of (2.32) the same computation does not vanish, so the misprint is detectable from within the example.

2. Equation (3.6), p. 64. The conjugate components are printed as

$$\bar{u} = \bar{u}_j e_j \quad \text{with} \quad u_j = a_j - ib_j.$$

Since (3.4) already sets $u_j = a_j + ib_j$, the left-hand side of the second relation should read $\bar{u}_j$. As printed, (3.4) and (3.6) together assert $a_j + ib_j = a_j - ib_j$.

3. Problem 9 of Chapter 3, p. 89. The problem asks the reader to show that, for the two-dimensional simple shear $\underline{F} = \underline{1} + \gamma \underline{e}_1 \otimes \underline{\hat{E}}_2$,

$$\underline{U} = \frac{1}{\sqrt{4 + \gamma^2}} \left[2\underline{\hat{I}} + \gamma(\underline{\hat{E}}_1 \otimes \underline{\hat{E}}_2 + \underline{\hat{E}}_2 \otimes \underline{\hat{E}}_1) + \gamma^2 \underline{\hat{E}}_1 \otimes \underline{\hat{E}}_1 \right].$$

The last dyad should be $\underline{\hat{E}}_2 \otimes \underline{\hat{E}}_2$. With the printed version, in matrix form $\underline{U} = \frac{1}{I} \begin{pmatrix} 2+\gamma^2 & \gamma \\ \gamma & 2 \end{pmatrix}$ with $I = \sqrt{4 + \gamma^2}$, one gets

$$\underline{U}^2 = \begin{pmatrix} 1 + \gamma^2 & \gamma \\ \gamma & 1 \end{pmatrix},$$

which is the matrix of $\underline{B}$, not of $\underline{C}$; the required $\underline{U}^2 = \underline{C} = \begin{pmatrix} 1 & \gamma \\ \gamma & 1 + \gamma^2 \end{pmatrix}$ of (3.55) is obtained only with $\gamma^2 \underline{\hat{E}}_2 \otimes \underline{\hat{E}}_2$:

$$\underline{U}^2 = \frac{1}{4 + \gamma^2} \begin{pmatrix} 4 + \gamma^2 & \gamma(4 + \gamma^2) \\ \gamma(4 + \gamma^2) & (4 + \gamma^2)(1 + \gamma^2) \end{pmatrix} = \begin{pmatrix} 1 & \gamma \\ \gamma & 1 + \gamma^2 \end{pmatrix},$$

the last entry using $\gamma^2 + (2 + \gamma^2)^2 = 4 + 5\gamma^2 + \gamma^4 = (4 + \gamma^2)(1 + \gamma^2)$.

The printed expression is in fact the correct formula for the *left* stretch $\underline{V}$, with hats attached: applying the same identity $I\underline{V} = \underline{B} + J\underline{I}$ to (3.56) gives

$$\underline{V} = \frac{1}{\sqrt{4 + \gamma^2}} \left[2\underline{\hat{I}} + \gamma(\underline{e}_1 \otimes \underline{e}_2 + \underline{e}_2 \otimes \underline{e}_1) + \gamma^2 \underline{e}_1 \otimes \underline{e}_1 \right],$$

so the two seem to have been interchanged. The distinction is exactly the one the text itself stresses between (3.53) and (3.54), where the coefficient $1 + \gamma^2$ sits on $\underline{\hat{E}}_2 \otimes \underline{\hat{E}}_2$ in $\underline{C}$ and on $\underline{e}_1 \otimes \underline{e}_1$ in $\underline{B}$.

4. The component form of $\underline{L}$ in the spinning disc, p. 96. Immediately below (4.27) the text writes “Thus, $L_{ij} = \omega e_{ijk} n_j$ ” and, two lines further on, “ $W_{ij} = \omega e_{ijk} n_j$ ”. Neither can stand as printed: the index j appears three times, once free on the left and twice on the right, so the equations are not index balanced, and k is left free on the right with no counterpart on the left.

Equation (4.27) itself,

$$v_i(\underline{y}, t) - v_i(\underline{x}, t) = \omega e_{ijk} n_j (y_k - x_k),$$

is correct. Comparing it with $v_i(\underline{y}, t) - v_i(\underline{x}, t) = L_{ik} (y_k - x_k)$ identifies i and k as the free indices and j as the summed one, so the two lines should read

$$L_{ik} = \omega e_{ijk} n_j, \quad W_{ik} = \omega e_{ijk} n_j.$$

The intervening conclusion is then immediate, $L_{ki} = \omega e_{kji} n_j = -\omega e_{ijk} n_j = -L_{ik}$, so $D_{ik} = 0$ and the motion is rigid, as the text says. Nothing downstream is affected.

A second, smaller slip occurs in the chain (4.28). Its second line is printed as $\frac{1}{2} \omega e_{jki} L_{ik} \underline{e}_j$ and its third as $\frac{1}{2} \omega e_{jki} e_{ilk} n_l \underline{e}_j$, so the factor ω appears in both. Since L_{ik} already carries ω , the factor should enter exactly once, at the step where L is substituted:

$$\underline{w} = \frac{1}{2} e_{jki} v_{i,k} \underline{e}_j = \frac{1}{2} e_{jki} L_{ik} \underline{e}_j = \frac{1}{2} \omega e_{jki} e_{ilk} n_l \underline{e}_j = \frac{1}{2} \omega (2\delta_{jl}) n_l \underline{e}_j = \omega \underline{n}.$$

The stated result $w = \omega \underline{n}$ is of course correct.

5. A sign in equation (6.116), p. 134. The printed equation reads

$$e_{ijk}x_j\{\rho(\dot{v}_k - b_k) + T_{kl,l}\} = e_{ijk}T_{kj}.$$

The sign inside the braces should be negative. Expanding $\rho\dot{\psi}_i = \rho r_i + A_{il,l}$ with the identifications (6.114),

$$\rho e_{ijk}x_j\dot{v}_k = \rho e_{ijk}x_j b_k + e_{ijk}T_{kj} + e_{ijk}x_j T_{kl,l},$$

and collecting gives $e_{ijk}x_j\{\rho(\dot{v}_k - b_k) - T_{kl,l}\} = e_{ijk}T_{kj}$.

The sentence immediately after the equation requires the minus sign as well: it says the braces vanish by (6.110). Since (6.110) reads $\rho\dot{v}_k = \rho b_k + T_{kl,l}$, the combination that vanishes is $\rho(\dot{v}_k - b_k) - T_{kl,l}$. With the printed plus sign the braces would equal $2T_{kl,l}$ on solutions of (6.110), and the symmetry of the Cauchy stress would not follow.

6. A missing transpose in equation (9.14), p. 183. The second term of the printed (9.14) is $\dot{Q}_1 Q_1^t(\underline{c}_1 - \underline{c}_2)$. It should be $\dot{Q}_1^t(\underline{c}_1 - \underline{c}_2)$.

The error is visible without computing anything, from the spaces the factors live in. The vectors $\underline{c}_k$ lie in $\mathbb{E}^{3+}$, since (9.6) adds them to $\underline{Q}_k x_k$ there. Now $\dot{Q}_1^t$ carries $\mathbb{E}^{3+}$ to $\mathbb{E}^3$ and $\dot{Q}_1$ carries $\mathbb{E}^3$ back to $\mathbb{E}^{3+}$, so the printed expression lands in $\mathbb{E}^{3+}$. But the vector it is added to must lie in $\mathbb{E}^3$, because (9.13) adds it to $\underline{Q}v_1$ there. Exactly one transpose is missing.

A check of the corrected form: with $\underline{Q}_2 = \underline{Q}_1$ one has $\underline{Q} = \underline{I}$ and (9.12) reduces to a pure translation, for which the corrected term reproduces the relative velocity correctly while the printed one does not.

7. A repeated index in equation (9.35), p. 191. The determinant is printed as

$$(\eta_1 - \eta_1)(\eta_2 - \eta_3)(\eta_3 - \eta_1),$$

whose first factor is identically zero. The intended factor is $\eta_1 - \eta_2$, giving the Vandermonde form $(\eta_1 - \eta_2)(\eta_2 - \eta_3)(\eta_3 - \eta_1)$. The argument at that point requires the determinant to be non-zero when the three principal values are distinct, and with the printed factor it vanishes always, which would destroy the conclusion.

8. The 11 component of $\underline{B}^2$ in (11.37) and (11.39), p. 213–214. Both displays print the 11 component of $\underline{B}^2$ in simple shear as $(1 + \gamma^2) + \gamma^2$. It should be $(1 + \gamma^2)^2 + \gamma^2$. With

$$(B_{ij}) = \begin{pmatrix} 1 + \gamma^2 & \gamma & 0 \\ \gamma & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

squaring gives $B_{11}^2 = B_{11}B_{11} + B_{12}B_{21} = (1 + \gamma^2)^2 + \gamma^2$. At $\gamma = 1$ the correct value is 5 and the printed expression gives 3; direct squaring of $\begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ confirms 5. The other two entries of the printed $\underline{B}^2$, namely $1 + \gamma^2$ and $\gamma(2 + \gamma^2)$, are correct.

The consequence is not confined to that entry. Equation (11.42), $T_{11} - T_{22} = \gamma T_{12}$, is derived from these components and is described in the text as a universal relation furnishing a critical test of the theory. With the printed value it comes out as $T_{11} - T_{22} = \gamma^2(f_1 + f_2)$, which is not γT_{12} unless $f_2 = 0$. With the corrected value it comes out exactly.

9. A spurious factor γ in equation (11.40), p. 214. The printed shear response is

$$T_{12} = \gamma\mu(\gamma^2), \quad \mu(\gamma^2) = f_1(\gamma^2) + \gamma(2 + \gamma^2)f_2(\gamma^2).$$

The factor γ in the second term does not belong; the correct expression is $\mu(\gamma^2) = f_1(\gamma^2) + (2 + \gamma^2)f_2(\gamma^2)$.

Two independent checks. First, the 12 entry of (11.39) is $f_1\gamma + f_2\gamma(2 + \gamma^2)$, and factoring out the γ that (11.40) displays leaves $f_1 + (2 + \gamma^2)f_2$. Second, the notation $\mu(\gamma^2)$ asserts that μ depends on γ only through γ^2 , whereas the printed expression contains a term odd in γ and so contradicts its own argument list. The same spurious factor also breaks (11.42), which is derived from μ two lines later.

Part II. Statements that appear stronger or looser than intended

10. The sign of $\det \underline{F}$ and occupiability, p. 50. The text states: “If κ is not occupiable, then $\det \underline{F} < 0$ and (2.50)₂ is replaced by $J = |\det \underline{F}|$.”

A non-occupiable reference configuration does not by itself force $\det \underline{F} < 0$. What the preceding argument establishes is the converse direction: if κ is occupiable then $\det \underline{F} > 0$. A reference configuration may fail to be occupiable while still being reachable, at each material point, by an orientation-preserving map; the obstruction to occupiability is global, whereas the sign of $\det \underline{F}$ is a pointwise matter. Would “ $\det \underline{F}$ need not be positive, and (2.50)₂ is then replaced by $J = |\det \underline{F}|$ ” be the intended reading? If the stronger statement is intended, a word on why non-occupiability forces the orientation reversal would be very welcome, since this is the first place in the book where the reader is asked to accept a sign convention on faith.

11. Equation (2.9) and the term “material velocity”, p. 40. Equation (2.9) introduces $\tilde{v}(x, t) = v(\chi^{-1}(x, t), t)$ with the words “Alternatively, the material velocity is given by the function $\tilde{v}$ ”. In §2.2, however, the tilde is reserved for the *spatial* description, and $\tilde{v}$ is precisely the spatial description of the velocity field. Since §2.2 goes to some length to insist that $\bar{f}$, $\tilde{f}$, $\hat{f}$ are three different functions with three different domains, calling $\tilde{v}$ “the material velocity” one page earlier is likely to be read as a contradiction. Is “the velocity in the spatial description” intended? The same phrase recurs after (2.17).

12. Equation (3.19) and the labels λ_1, λ_2 , p. 66. In §3.2, λ_1 and λ_2 are defined as the minimum and maximum of $f(\underline{u}) = \underline{u} \cdot \underline{A}\underline{u}$ on the unit sphere, and $\underline{u}_1, \underline{u}_2$ as the corresponding minimizers and maximizers. Everywhere else in the book, including (3.15) and (3.38), the subscript on λ_i is a plain enumeration index over the three principal values. In a three-dimensional problem the extremal pair is the smallest and the largest, so the local convention of §3.2 makes λ_2 denote the *largest* principal value while the global convention leaves it as the middle one. Would $\lambda_{\min}, \lambda_{\max}$ (or λ_-, λ_+) in §3.2 avoid the clash? Equation (3.26) is where the two conventions are hardest to hold apart.

13. Problem 21(g), Chapter 1, p. 35. The identity $\det(\underline{A} + \underline{a} \otimes \underline{b}) = (\det \underline{A})(1 + \underline{a} \cdot \underline{A}^{-t}\underline{b})$ presupposes that $\underline{A}$ is invertible, but the hypothesis is not stated, and parts (a)–(f) of the same problem carry no such restriction. Part (d) of the problem does state its hypothesis implicitly by writing $\underline{A}^{-1}$; part (g) does so only through the $\underline{A}^{-t}$ on the right-hand side, which a student is likely to reach only after having attempted the general case.

14. Equation (3.53), third line, p. 73. The cross terms are printed as $\gamma(\hat{\underline{E}}_2 \otimes \underline{1}^t \underline{E}_1 + \underline{1}^t \underline{E}_1 \otimes \hat{\underline{E}}_2)$. Expanding $(\underline{1} + \gamma \underline{e}_1 \otimes \hat{\underline{E}}_2)^t (\underline{1} + \gamma \underline{e}_1 \otimes \hat{\underline{E}}_2)$ by the rules of Problem 17 produces

$\underline{1}^t \underline{e}_1$, not $\underline{1}^t \underline{E}_1$: the vector being shifted back is the *spatial* basis vector $\underline{e}_1$ appearing in (3.51). The two agree only after the identification $\{\underline{E}_A\} = \{\underline{e}_i\}$ made at (3.49), and the passage to the fourth line, $\underline{1}^t \underline{E}_1 = \hat{\underline{E}}_1$, uses (2.68) directly and therefore really does want $\underline{E}_1$. Since §2.6 is at pains to explain that $\underline{e}_i$ and $\underline{E}_A$ live in the same space but are not the same vectors in general, could the identification be recalled explicitly at this step? As it stands, the example silently uses a convention adopted four equations earlier for an unrelated purpose.

15. Equation (3.57), p. 74. The chain $\det \underline{U} = [\underline{U}u_1, \underline{U}u_2, \underline{U}u_3]/[u_1, u_2, u_3] = [\lambda_1 u_1, \lambda_2 u_2, \lambda_3 u_3] = \lambda_1 \lambda_2 \lambda_3$ drops the denominator between the second and third members. The step is of course legitimate, since $[u_1, u_2, u_3] = \pm 1$ for an orthonormal triad and the sign cancels against the numerator, but as printed the equality signs do not hold term by term.

Part III. Notation and points where a clarifying sentence would help

16. The symbol e for both basis vectors and the permutation symbol, Chapter 1. In (1.164) the single line

$$\underline{e}_j \cdot \text{curl } \underline{y} = \text{div}(\underline{v}_i \underline{e}_i \times \underline{e}_j) = \text{div}(\underline{v}_i e_{ijk} \underline{e}_k) = (\underline{v}_i e_{ijk})_{,k} = e_{kij} v_{i,k}$$

uses e as a basis vector and as the alternating symbol in the same product. The distinction is carried entirely by the boldface, which at the size used for subscripted symbols is easy to lose. Many texts use ε_{ijk} for the permutation symbol for exactly this reason. Is there a reason for the present choice that I have missed? The same collision occurs in Problem 11(a) of Chapter 1 and in Problem 1 of Chapter 2.

17. The symbol δ_{iA} in (2.70), p. 54. The shifter is decomposed as $\underline{1} = \delta_{iA} \underline{e}_i \otimes \hat{\underline{E}}_A$, and the text then observes that “ δ_{iA} are the components of the usual Kronecker delta if, and only if, $\{\underline{E}_A\} = \{\underline{e}_i\}$. Otherwise, they are the cosines of the fixed angles between the elements of the two sets.”

This is clear once read, but the notation works against the sentence: the symbol δ is used throughout the book with the meaning $\delta_{ij} = \underline{e}_i \cdot \underline{e}_j$, and (2.74) then reads $\delta_{iA} \delta_{iB} = \delta_{AB}$, in which the two δ ’s on the left have the general meaning and the one on the right has the Kronecker meaning. Would a distinct symbol for the shifter components, s_{iA} say, be worth the extra letter? This is the one place in Chapter 2 where I had to reread several times to be sure which object was which.

18. Equation (2.12), p. 40: the absence of the subscript κ . I record this as a query answered rather than as a correction, since it is the first thing a reader checks. Equation (2.12) reads $\tilde{f}(\underline{x}, t) = \tilde{f}(\underline{\chi}^{-1}(\underline{x}, t), t)$ with no subscript on $\underline{\chi}$, whereas the very next equation, (2.13), carries $\underline{\chi}_\kappa$ throughout, and the reader naturally suspects an omission. It is not one: $\underline{\chi}^{-1}$ in (2.12) is the inverse of the motion (2.1), which returns the *material point* p , and $\tilde{f}$ is precisely the function that eats p , so the unsubscripted map is the correct one; $\underline{\chi}_\kappa^{-1}$ would return $\underline{X}$ and would have to be composed with $\hat{f}$ instead. The same applies to (2.9).

The point is worth a half-sentence in the text, because $\underline{\chi}$ and $\underline{\chi}_\kappa$ differ by the fixed bijection κ alone and the eye does not distinguish them; the reader who assumes a misprint here will misread the entire section. A parenthetical “(note that (2.1) rather than (2.6) is inverted here, since $\tilde{f}$ takes the material point as its argument)” would settle it.

19. $\underline{x}' = \underline{0}$ in the sanity check preceding §2.4, p. 44. The line “application of (2.20) to the position vector $\underline{x}$, combined with $\underline{x}' = \underline{0}$, yields $\dot{x}_i = x_{i,j}\tilde{v}_j = \delta_{ij}\tilde{v}_j = \tilde{v}_i$, as expected” is correct but is delivered without comment, and $\underline{x}' = \underline{0}$ reads as a flat contradiction of the fact that material points move. The content is that the field in question, taken in the spatial description, is $\tilde{u}(\underline{x}, t) = \underline{x}$, the identity map of $\mathcal{E}^3$, which carries no t at all; the motion resides entirely in $\hat{u} = \chi_\kappa$. Given how much care §2.2 devotes to the distinction between $'$ and $\cdot$, and given that this is the extreme illustration of it ($\underline{0}$ against $\underline{v}$ for the same field), one line of explanation here would repay itself. As printed, “as expected” invites the reader to believe the step is routine when in fact it is the sharpest test of whether §2.2 has been understood.

20. Section title of §2.2 versus the order in (2.11). The section is titled “Material, referential and spatial descriptions”, whereas (2.11) lists $\bar{f}$, $\tilde{f}$, $\hat{f}$ and the sentence after (2.13) reads “the material, spatial and referential descriptions, respectively”. The orders differ. A trivial matter, but the reader who uses the section title as a mnemonic for which decoration is which will be misled.

21. Equation (2.96), p. 58. The step $\frac{1}{2}(\underline{M} \cdot \underline{C} \underline{M} - 1) = \underline{M} \cdot \underline{E} \underline{M}$ uses $1 = \underline{M} \cdot \hat{\underline{I}} \underline{M}$, which holds because $|\underline{M}| = 1$. The unit-length hypothesis is stated at (2.66) but is not recalled here, and the reader tracking the algebra will look for a missing factor. Similarly at (2.98).

Part IV. Figures

22. Figure 2.5, p. 41. The caption reads “Different material points occupy position $\underline{x}$ at different times”, but the figure shows two disjoint regions. Since the whole point of the spatial description is that *one* body sweeps past *one* fixed place, a single body drawn at two instants, with a single marked point $\underline{x}$ held fixed and two different material points successively coinciding with it, would carry the idea considerably better. As drawn, the reader can come away believing that $\underline{x}$ is a point of each region rather than a point of space.

23. Figure 2.9, p. 50. In the rendering of the deformation of oriented surface measure, the normals $\underline{N}$ and $\underline{n}$ read as though they lie in the respective tangent planes rather than perpendicular to them; the parallelograms spanned by $d\underline{X}, d\underline{Y}$ and $d\underline{x}, d\underline{y}$ are drawn nearly edge-on while the normals are drawn in the plane of the page. A perspective view in which the parallelogram is foreshortened and the normal clearly leaves the plane would remove the ambiguity. This figure carries the Piola–Nanson formula, which is one of the results students most often get the geometry of wrong.

24. Figure 2.8, p. 49. The three material curves intersecting at a point are shown with a parallelepiped whose hidden edges are omitted rather than dashed, which makes the orientation of the triad $d\underline{X}, d\underline{Y}, d\underline{Z}$ ambiguous, and with it the sign of $[d\underline{X}, d\underline{Y}, d\underline{Z}]$ in (2.48). Since the entire discussion that follows turns on that sign being positive, the handedness ought to be unmistakable from the figure.

Summary

Items 1 to 9 are, I believe, straightforward misprints and are verified above by independent computation. Item 10 is a mathematical statement that appears stronger than the argument supports. Items 11–15 are places where the printed equalities or terminology do not quite

hold as written, though the intended meaning is recoverable. Items 16–21 are requests for a clarifying phrase at points where a careful reader is likely to stall, and items 22–24 concern figures whose geometry conflicts with, or fails to support, the accompanying text.